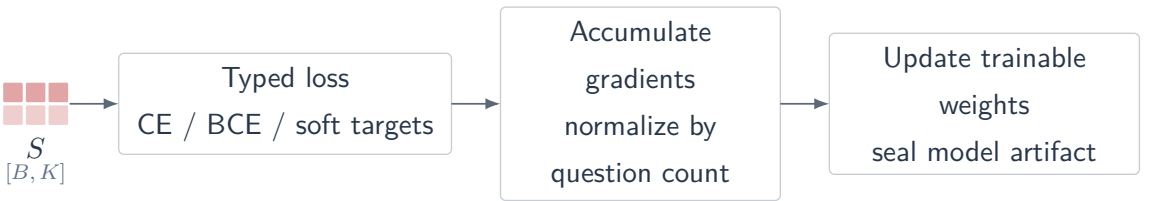
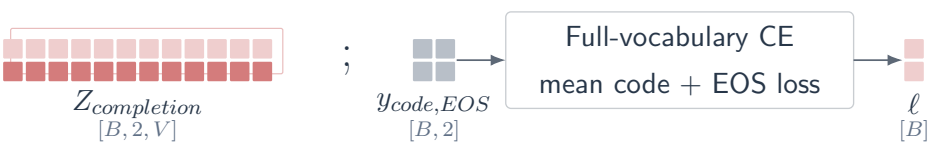
$$\mathcal{L} = \frac{1}{N}\sum_{b} \ell_b, \quad \tau = \exp(a), \quad p = \text{normalize}((S - \Delta)/\tau)$$

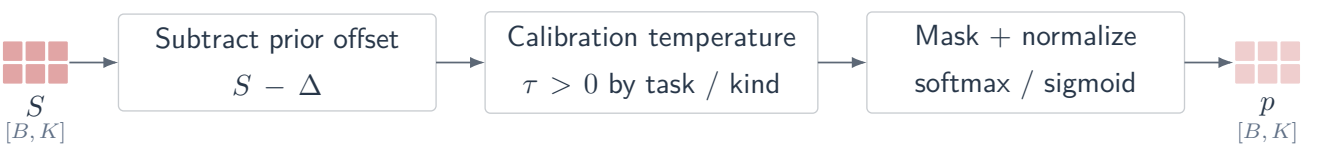
Train only on the training split; aggregate per-question losses



Completion training: two target positions retain the full vocabulary axis



Fit temperature on independent calibration data; apply the same transform in serving



Axes $Z_{completion} : [B, 2, V]$; targets $y : [B, 2]$ contain answer code and EOS. N is the number of questions contributing to an optimizer update. Δ is the configured prior offset; τ is temperature.

Meaning Training, calibration and evaluation splits have separate purposes. Calibration optimizes a probability objective, not necessarily accuracy, and can worsen held-out NLL. Offline and HTTP use the same task/kind routing and recorded fallback decisions.

Cell counts illustrate axes, not model sizes; tint identifies roles, not numeric values.